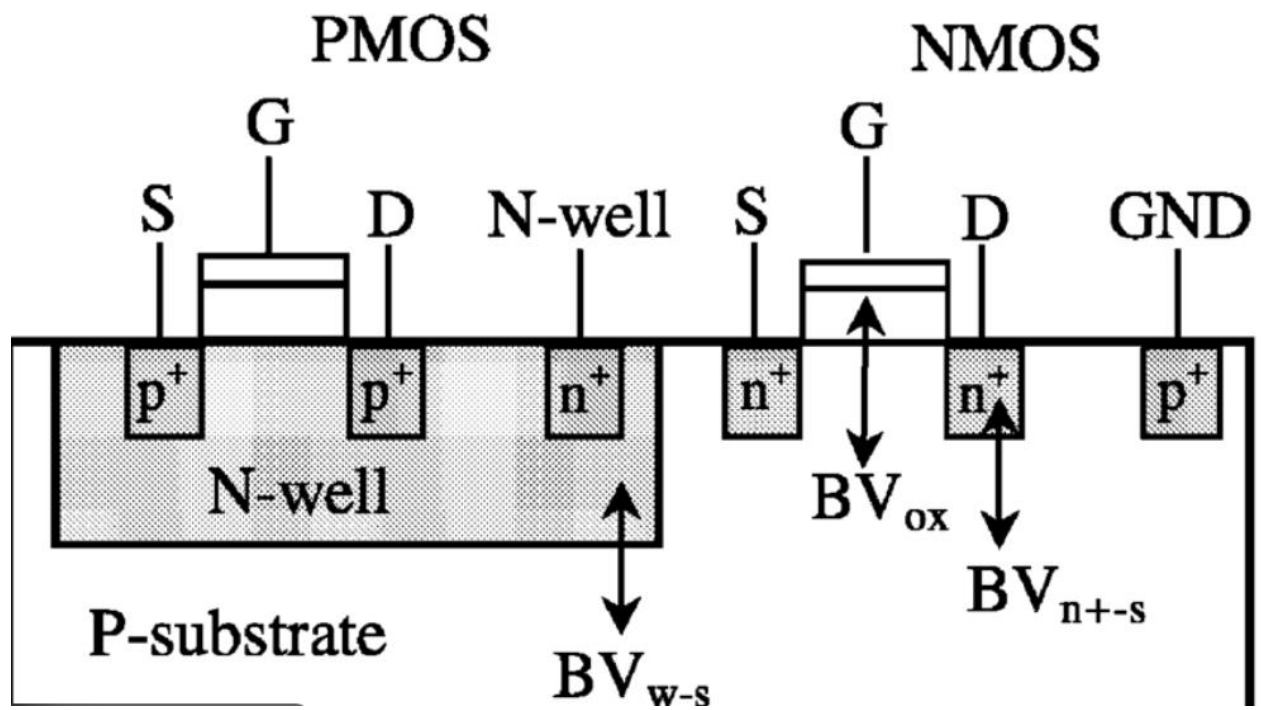
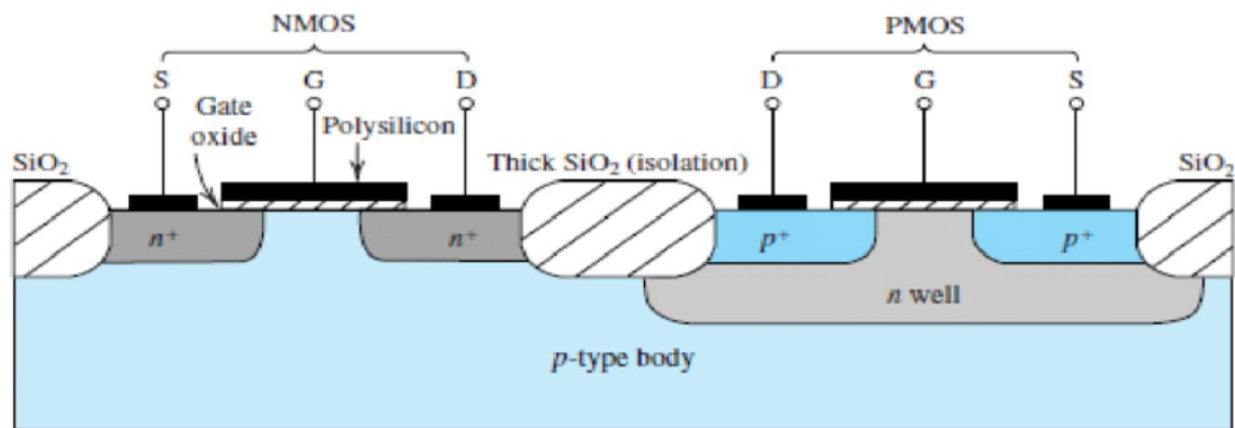
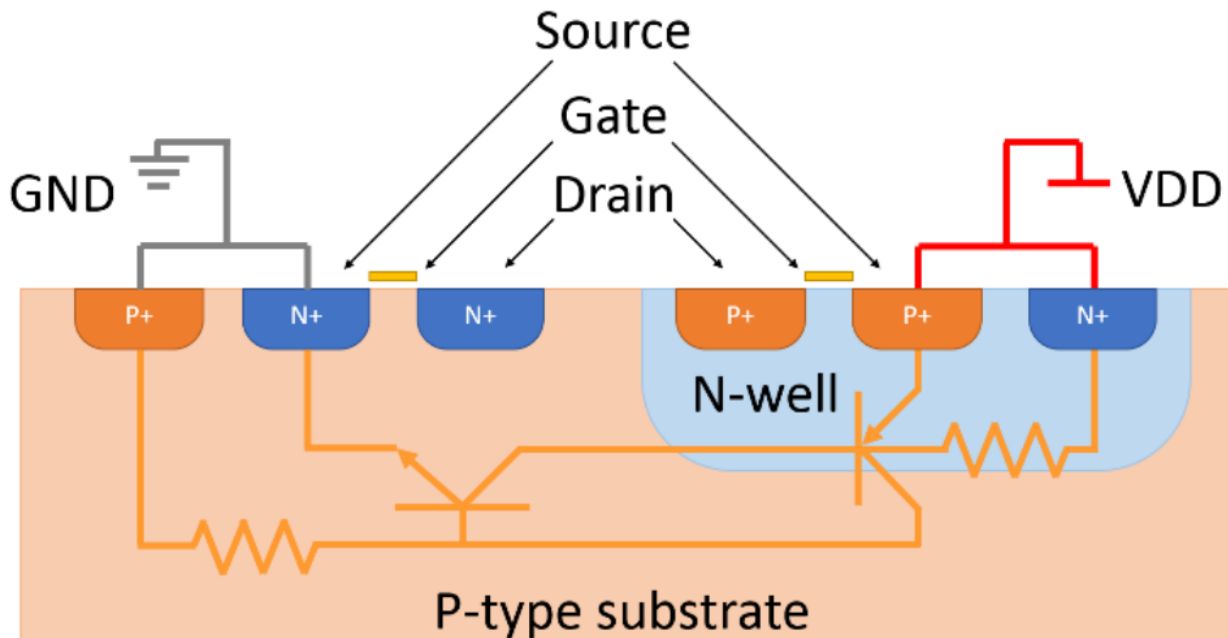


NANENG 301 PROJECT: PHASE 1

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1. CMOS: Basic Structure and Principles





What is CMOS, and why “complementary”?

- **Definition:** CMOS stands for *Complementary Metal–Oxide–Semiconductor*. It is a technology in which both NMOS (n-channel MOSFET) and PMOS (p-channel MOSFET) devices coexist on the same integrated-circuit silicon chip.

Complementary & symmetric: The “complementary” nature comes from using both n-channel and p-channel transistors: for a logic gate (inverter), one transistor pulls output high while the other pulls it low but ideally **only one is conducting at a time**, which leads to *very low static power consumption*.

Why CMOS dominates: Because of low standby power and scalability, CMOS became the standard technology for very largescale integration (VLSI).

Basic structural elements (cross-section view)

In a simplified CMOS structure you typically have:

- A **silicon wafer** (substrate), usually doped p-type (for standard n-well CMOS).

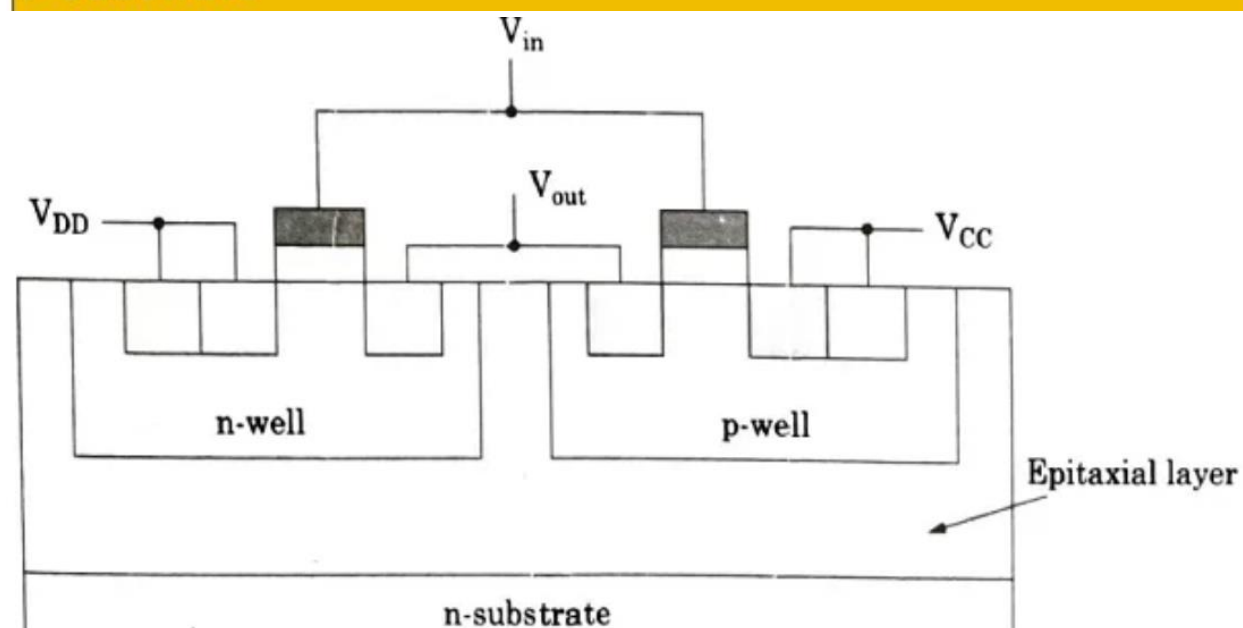
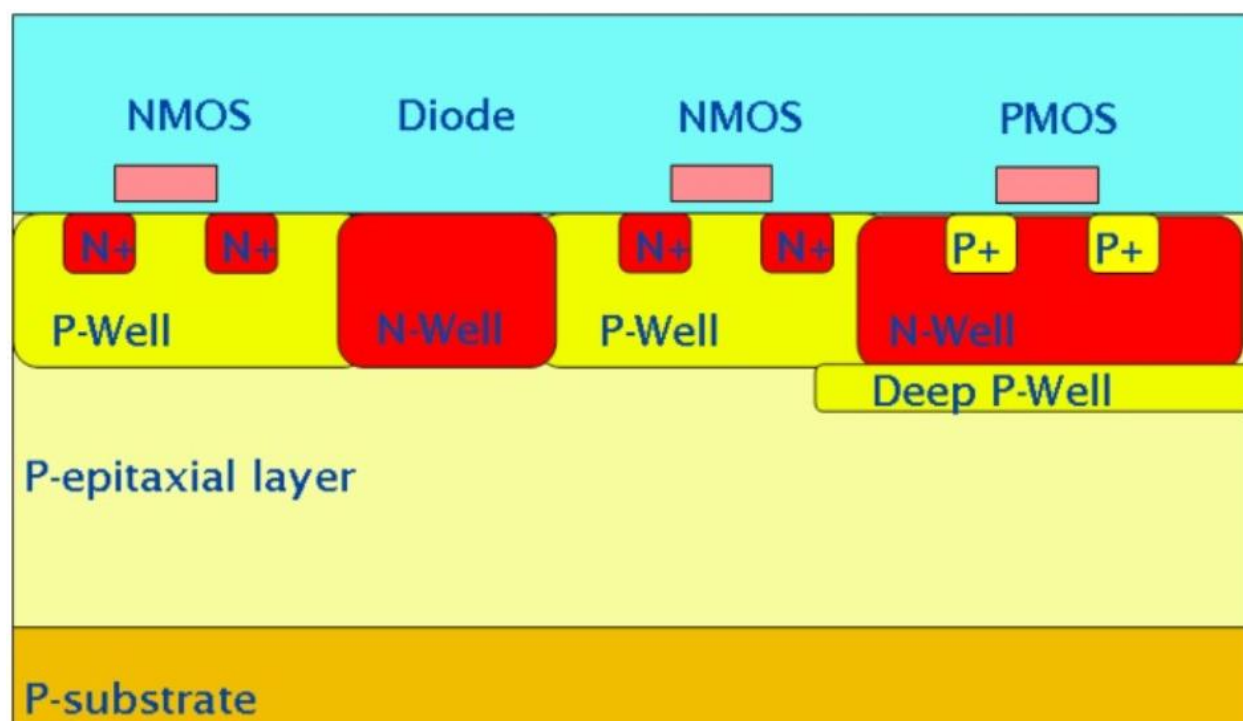
An **n-well** region implanted into the p-type substrate: this n-well hosts the PMOS transistors.

- **NMOS transistors** built directly in the p-type substrate (outside the n-well).
- **PMOS transistors** built inside the n-well (so that their body is n-type, opposite to p-type substrate).

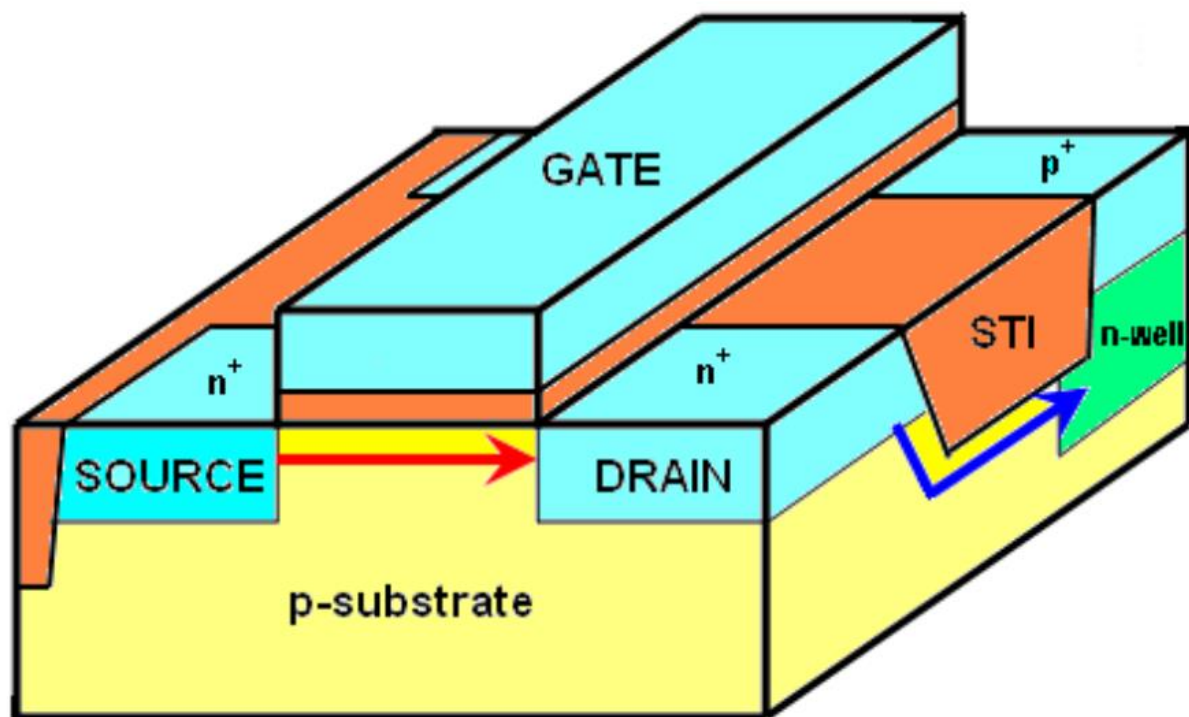
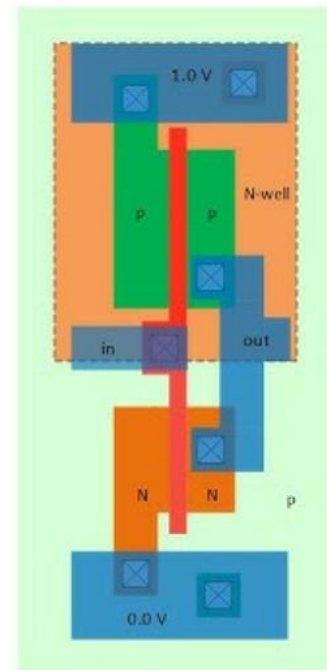
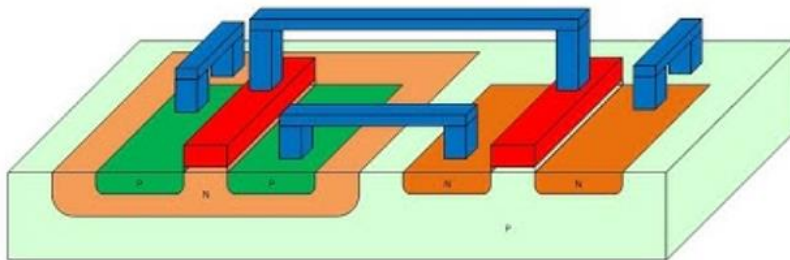
- **Gate oxide (SiO_2)** — a thin insulating oxide between the gate electrode (often polysilicon) and the silicon substrate. This forms the MOSFET gate.
- **Source and drain diffusion regions** (n^+ for NMOS, p^+ for PMOS), implanted into substrate (or well) adjacent to the gate, forming the transistor channels when gate bias is applied.
-
- **Isolation regions** (field oxide or trench/LOCOS zones) between devices to prevent unwanted conduction or interference between adjacent transistors.
-
- An example cross-section: NMOS in p-substrate, PMOS in n-well, gates on top (poly), source/drain diffusions to left/right, with SiO_2 isolation/field oxide around.
-

1.1 How NMOS and PMOS Are Fabricated on the Same Substrate (n-well / twin-well CMOS)

How do we get both NMOS and PMOS on the same wafer, given that they require different substrate doping (p-type for NMOS, n-type for PMOS)? The answer lies in well (tub) formation.



CMOS Technology: Layout



Well / Tub concept

- In a typical **n-well CMOS process**, you start with a **p-type silicon substrate**.
-

- Then, you **implant or diffuse n-type dopants** into selected regions to form **n-wells**. Within these n-wells, you build your **PMOS** devices. Outside the well (i.e., the original p-substrate), you build the **NMOS** devices.
-
- There is also “**twin-well**” (or **twin-tub**) CMOS, where both n-well and p-well are used (i.e., you can create separate p-well regions on an originally n-type substrate, or vice versa), allowing process flexibility or optimizing for certain device types.

Why this works electrically

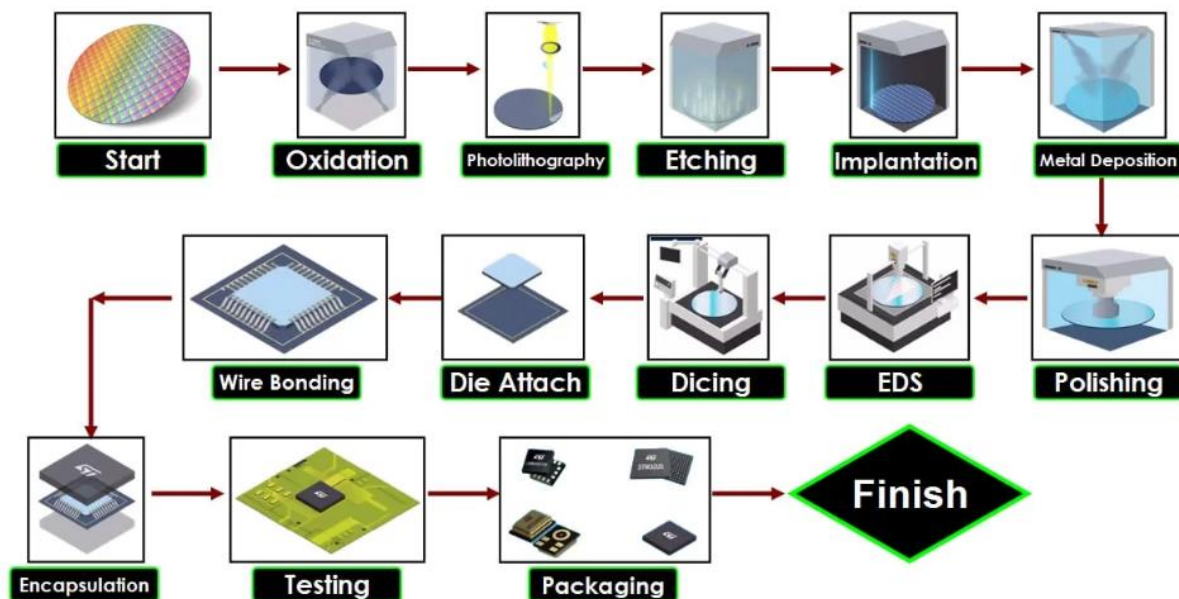
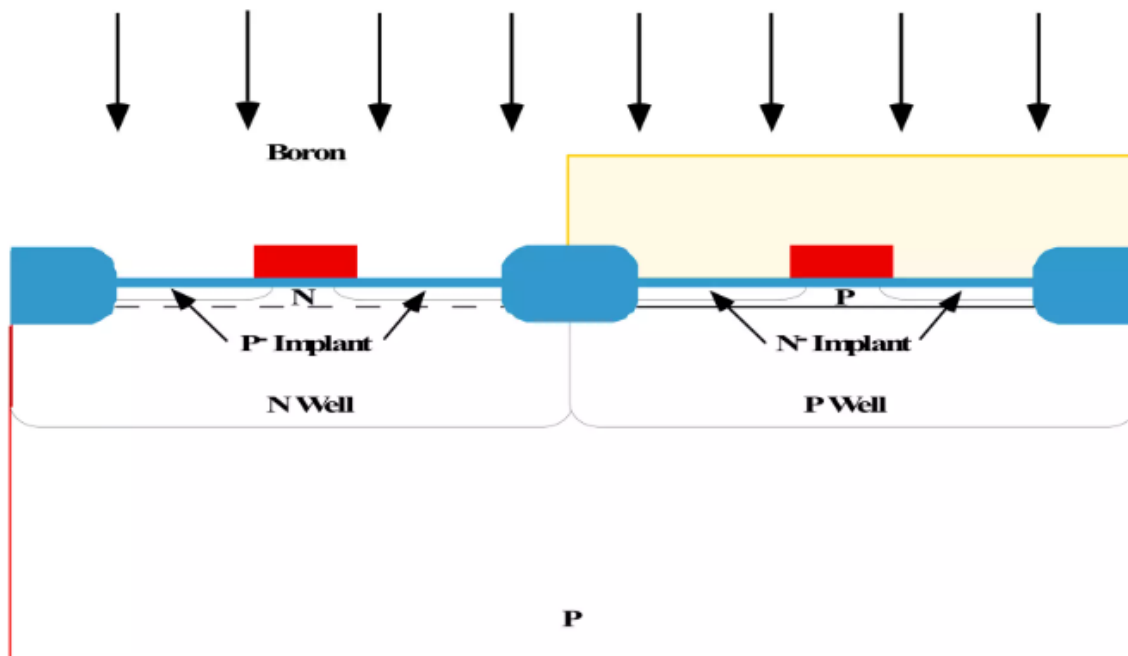
- Each MOSFET (NMOS or PMOS) essentially sees a correct “body” (bulk) potential: NMOS body is p-type substrate tied to lowest potential (VSS/ground), PMOS body is n-well tied to the highest potential (VDD). This ensures proper threshold behavior and avoids forward-biasing the body-source/junctions.
-
- The isolation (field oxide or wells) prevents the wells or substrate from interfering, so each transistor behaves independently.

This is how both transistor types live “side-by-side” on a *single* silicon wafer yet keep their intrinsic doping and biasing conditions correct.

1.2 Full CMOS Fabrication Sequence Step-by-Step Process Flow

Here I outline a typical, simplified CMOS fabrication flow (bulk CMOS, n-well process), from a bare wafer to finished devices. Real foundry flows are far more complex, but this covers the foundational steps. See the diagrams below for cross-section evolution.

CMOS Process Flow



Process

Thermal oxidation ($\text{Si} \rightarrow \text{SiO}_2$)

Photolithography

Etching

Purpose

Grow insulating oxide (gate oxide, field oxide)

Define patterns on the wafer using masks & photoresist

Remove material (oxide, poly, metal) selectively

Ion implantation / doping / diffusion	Introduce dopants to create wells, source/drain, tubs
Deposition (CVD/PVD)	Deposit materials: polysilicon, oxide, nitride, metal layers
Planarization (. CMP)	Flatten steps/bumps to enable multi-level metallization
Metallization & contacts	Create metal interconnects and vias to form circuits

Example n-well CMOS Flow (simplified)

Here is a **simplified process flow** for a CMOS inverter (i.e. one NMOS + one PMOS). Real chips may use many more masks and layers, but this illustrates the basics. This flow is consistent with standard references.

1. **Start with a clean silicon wafer**
 - a. Typically, a p-type silicon wafer is used for n-well CMOS.
 - b. The wafer is polished, cleaned, and prepared.
2. **Define isolation regions (active area) / Field oxide (Device isolation)**
 - a. A thin oxide + mask + nitride (or other hard-mask) is used so that only active device areas are protected; other areas will be oxidized.
 - b. Then perform a thick oxide growth (e.g. via LOCOS) or use trenches to form **field oxide**, isolating neighboring devices. Also channel-stop implants may be used to suppress parasitic conduction.
 - c. Result: “active regions” (where transistors will be) separated by thick field oxide.
3. **Form wells (. n-well)**
 - a. Use a **photolithography mask** to expose regions where wells are needed (for PMOS).
 - b. Perform **ion implantation (or diffusion)** of n-type dopants (like phosphorus or arsenic) into those exposed regions, forming n-well inside the p-substrate.
 - c. Anneal / drive-in to activate and diffuse dopants to desired depth/ profile.
4. **Grow or deposit gate oxide (thin SiO₂)** over the active areas (both NMOS and PMOS regions). This oxide is very thin (tens of nanometers), used later as the MOS gate dielectric.
5. **Deposit gate material** — typically **polysilicon** (poly-Si) — over the wafer. Then pattern it with photolithography + etch to form the **gates** of all transistors.
 - a. This “self-aligned gate” approach allows the gate to be used later as a mask for doping source/drain, ensuring alignment of source/drain to the gate → reduces parasitic capacitance and improves performance.
6. **Source/Drain doping (n⁺ for NMOS, p⁺ for PMOS)**
 - a. Use additional photolithography masks to define source/drain regions for NMOS and PMOS (active areas outside/inside well respectively).

- b. Perform **ion implantation**: n-type dopants for NMOS S/D, p-type for PMOS S/D.
 - c. Use the gate as a mask (“self-aligned”) so S and D align automatically with the gate edge.
 - d. After implantation, **anneal (heat)** to repair crystal damage and activate dopants.
- 7. Form body (well/substrate) contacts (taps)**
- a. To ensure proper biasing, heavy doped regions (p^+ under p-substrate or n^+ in well) are created for body contacts. These prevent floating wells and ensure stable transistor operation.
- 8. Deposit inter-layer insulator (oxide, nitride, etc.) and open contact windows (vias)**
- a. Cover the wafer with dielectric (oxide or nitride) to isolate gate/polysilicon and future metal layers.
 - b. Use photolithography + etch to create contact openings (windows) for source, drain, well contacts, and gate.
- 9. Metallization (first metal layer Al, Cu)**
- a. Deposit a metal layer over the wafer, then pattern it with photolithography + etch to form the metal interconnects connecting different transistor terminals (gates, sources, drains) as per circuit layout.
- 10. (Optional) Additional metal layers, vias, passivation, packaging**
- a. For real ICs, multiple metal layers (M1, M2, ...), vias, dielectric layers, and final passivation are applied. Then the wafer is diced into individual dies and packaged.

So in a highly simplified mask-count view (for a minimal CMOS inverter), you might have ~7 mask/processing steps: well formation, channel-stop/isolation, gate definition (poly), NMOS S/D doping, PMOS S/D doping, contact opening, metal patterning.

2. CMOS Technology Node Parameters

Year	Technology Node(nm)	Physical Gate(nm)	tox (nm)	Dielectric K	Vdd (V)	Vth (V)	Na (/cm ³)	Nd (/cm ³)	xj (nm)
2001	130	90	3.0	3.7	1.2	0.34	1.0e16	1.0e19	67.5
2004	90	53	2.4	3.0	1.1	0.32	1.4e16	1.4e19	46.7
2007	65	32	1.7	2.5	0.9	0.29	2.0e16	2.0e19	33.8
2010	45	22	1.5	2.0	0.8	0.29	2.9e16	2.9e19	23.4
2013	32	16	1.4	1.9	0.7	0.25	4.0e16	4.0e19	16.6
2016	22	11	1.3	1.7	0.6	0.22	5.9e16	5.9e19	11.4

SOURCE: R. Saleh, "Design and Technology Trends," University of British Columbia, Lecture Slides, 2008.

2.1 Gate Oxide Thickness (t_{ox})

Gate oxide is the thin insulating layer between the gate and the silicon channel, and its thickness directly affects how strongly the gate controls the channel. This relationship is described by the oxide capacitance:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}}.$$

As CMOS technology advances, the oxide becomes thinner so that the gate can better modulate the inversion charge. This thinning also appears in the threshold voltage equation:

$$V_{TH} = V_{FB} + 2\phi_F + \frac{\sqrt{2\epsilon_s q N_A (2\phi_F)}}{C_{ox}}.$$

A thinner oxide (larger C_{ox}) reduces the depletion term in this expression and gives stronger electrostatic control. Across the nodes examined, the oxide thickness decreases from about **3 nm** at 130 nm technology to **just above 1 nm** at the 22 nm node. At these extreme thicknesses, direct tunneling becomes significant, so advanced nodes use high- κ materials instead of pure SiO₂.

2.2 Junction Depth

Junction depth refers to how deeply the source and drain dopants extend below the silicon surface. This distance is crucial because shallower junctions reduce short-channel effects and lower parasitic capacitances.

Early technology nodes formed junctions mainly through diffusion, which spreads dopants according to:

$$x_j = 2\sqrt{Dt} \operatorname{erfc}^{-1}\left(\frac{C_B}{C_s}\right)$$

Modern nodes rely instead on ion implantation, which creates a more controlled Gaussian profile:

$$N(x) = N_p \exp\left[-\frac{(x - R_p)^2}{2\Delta R_p^2}\right]$$

As technology scales, junction depths shrink from around **70 nm** at 130 nm down to **about 10 nm** at the 22 nm node. Low-energy implantation and extremely fast thermal annealing steps are used to keep these junctions shallow and well-defined.

2.3 Channel Doping Concentration

The channel (or well) doping sets the charge distribution under the gate and plays a major role in tuning the threshold voltage. The relationship appears directly in the threshold voltage equation through the term:

$$\sqrt{2\epsilon_s q N_A (2\phi_F)}$$

Higher channel doping reduces the depletion width and helps prevent punch-through as transistors get shorter. For this reason, the channel doping concentration N_A steadily increases with scaling, rising from the low 10^{16} per cubic centimeter range at the 130 nm node to nearly 6×10^{18} per cubic centimeter by the 22 nm node.

It is important to note that channel doping is not the same as substrate doping, nor is it the extremely high source/drain doping. Channel doping is a deliberately adjusted level in the well region where the device operates.

2.4 Substrate Doping Concentration

The substrate is the lightly doped silicon wafer on which fabrication begins. Its doping level is usually between 10^{15} and 10^{16} per cubic centimeter. This provides a clean, high resistivity starting point before the wells are implanted.

To avoid confusio

- Substrate doping = original wafer doping (light).
- Channel doping = modified doping in the active well region (higher).
- Source/drain doping = extremely high doping to reduce resistance (around 10^{19} per cubic centimeter).

Only channel doping (N_A) and source/drain doping (N_D) appear in typical scaling tables; the substrate doping is not listed but is known to remain lightly doped.

2.5 Source/Drain Doping Concentration

Source and drain regions require very high dopant concentrations to make good ohmic contacts and minimize parasitic resistances. These regions are formed by high-dose implantation, and their concentration profiles follow:

$$N(x) = N_p \exp\left(-\frac{(x - R_p)^2}{2\Delta R_p^2}\right)$$

The total implanted dose is approximately:

$$\phi \approx 2\pi\Delta R_p N_p$$

Typical values for N_p are around 10^{19} per cubic centimeter, increasing slightly with each technology node.

2.6 Summary of Technology Parameters

Node	t_{ox} (nm)	x_j (nm)	N_A (cm ⁻³)	N_d (cm ⁻³)	Substrate (cm ⁻³)
130 nm	3.0	67.5	1.0×10^{16}	1.0×10^{19}	10^{15} – 10^{16}
90 nm	2.4	46.7	1.4×10^{16}	1.4×10^{19}	10^{15} – 10^{16}
65 nm	1.7	33.8	2.0×10^{16}	2.0×10^{19}	10^{16}
45 nm	1.5	23.4	2.9×10^{16}	2.9×10^{19}	10^{16}
32 nm	1.4	16.6	4.0×10^{16}	4.0×10^{19}	10^{16}
22 nm	1.3	11.4	5.9×10^{16}	5.9×10^{19}	10^{16}

3. Investigate Process Dependencies

3.1 Gate Oxide Thickness (t_{ox}) [9]

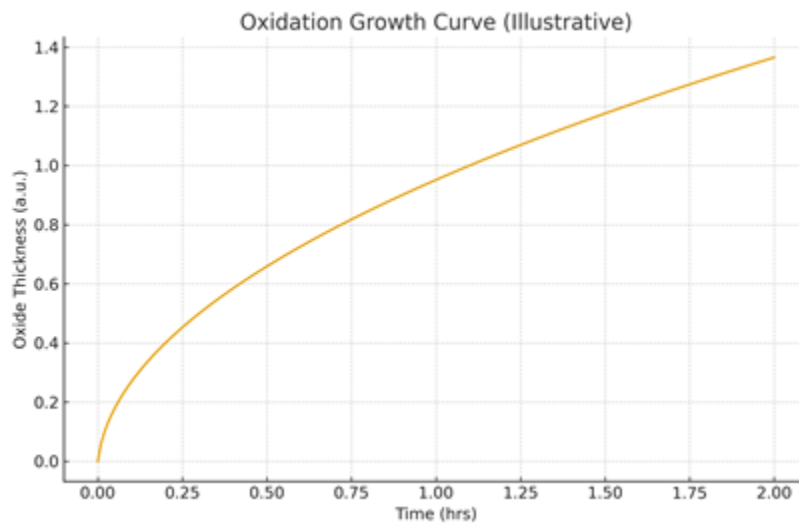
Gate oxide thickness influences gate control, capacitance, threshold voltage, leakage, and short-channel behavior. The oxide thickness is primarily set by thermal oxidation or high- κ deposition.

Governing Equation (Deal–Grove Model)

$$x_{\text{ox}}^2 + (A \cdot x_{\text{ox}}) = B (t + \tau)$$

Example Calculation

For a dry oxidation process at 1000°C with coefficients $A = 0.1$ and $B = 1.0$ (illustrative), solving the quadratic gives the oxide thickness as shown in Figure 1.



3.2 Junction Depth (x_j) [10]

Junction depth is controlled by implantation energy and dose, followed by diffusion during annealing. Shallower junctions improve short-channel behavior but increase resistance. [13]

Governing Equations

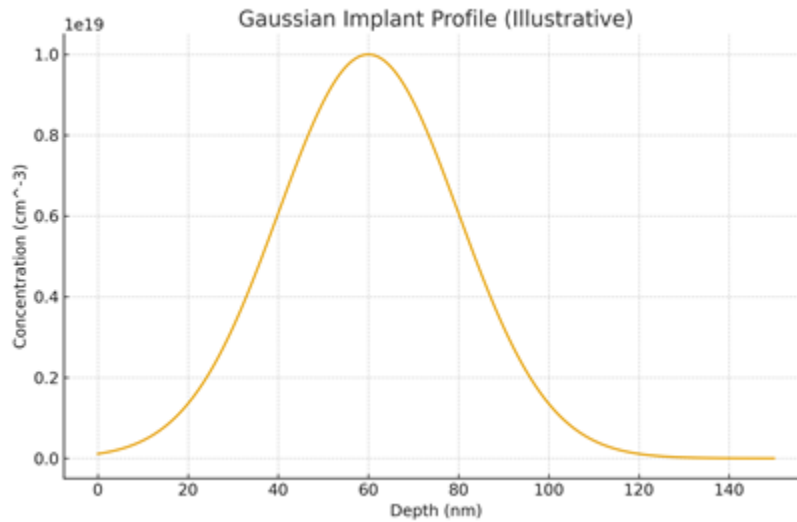
$$N(x) = N_p \cdot \exp(-(x - R)^2 / (2\Delta R_p^2))$$

$$x_j \approx \sqrt{(4D(T)t_{\text{anneal}})}$$

Example Calculation

Assuming an implant with $R_p = 60$ nm and $\Delta R_p = 20$ nm, the Gaussian distribution is shown below. Using a diffusivity $D(T) = 2 \times 10^{-14}$ cm²/s and anneal time of 10 s, the junction depth is approximately:

$$x_j \approx \sqrt{(4 \times 2 \times 10^{-14} \times 10)} \approx 2.8 \times 10^{-6} \text{ cm} = 28 \text{ nm}.$$



3.3 Channel Doping (N_{ch})

Channel doping sets the threshold voltage and affects depletion width and mobility. It is controlled by well implants, halo implants, and threshold-adjust implants. [12]

Governing Equation for Threshold Voltage

$$V_T = \phi_{ms} + 2\phi_f + \sqrt{(4q\epsilon_s N_A \phi_f) / C_{ox}}$$

Example Calculation

For a channel doping concentration $N_A = 1 \times 10^{16}$ cm⁻³, the Fermi potential is:

$\phi_f = (kT/q) \ln(N_A / n_i)$. At room temperature (300 K), $kT/q \approx 0.026$ V and $n_i \approx 1 \times 10^{10}$ cm⁻³, yielding:

$$\phi_f \approx 0.026 \ln(1 \times 10^6) \approx 0.36 \text{ V}.$$

This value directly contributes to the threshold voltage expression.

3.4 Substrate Doping (N_{sub})

Substrate doping determines bulk resistivity, depletion width, and body effect. It is primarily fixed by the wafer vendor but is modified locally by wells. [11]

Depletion Width Equation

$$W_d = \sqrt{2 \epsilon_s (2\phi_f) / (qN_A)}$$

Example Calculation

For $N_{sub} = 1 \times 10^{15} \text{ cm}^{-3}$ and $\phi_f = 0.36 \text{ V}$, assuming $\epsilon_s = 1.04 \times 10^{-12} \text{ F/cm}$,

$$W_d \approx \sqrt{2 \times 1.04 \times 10^{-12} \times 0.72 / (1.6 \times 10^{-19} \times 1 \times 10^{15})} \approx 0.32 \text{ } \mu\text{m}.$$

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